

# Relay Function Transparent Transmission Function Instruction Manual

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## Part I Overview

All of our **Supreme Edition** products are equipped with relay and transparent transmission function. The so-called relay is to connect two CAN bus networks running at different (or the same) baud rates to each other through the two channels of the CAN adapter to realize data sharing, which is similar to the function of "network relay".

This document focuses on the functions of relaying, transparent transmission and how to use the relaying software.

## Part II Relay, Transparent Transmission Functions

### 2.1 Relay

The USB-CAN adapter and the **Supreme Edition of CANalyst-II** analyzer products provided by our company support the CAN relay function, which has the following features:

① Realize data exchange between 2 CAN channels at the same or different baud rates. (Only need to be configured on the PC side, can be detached from the PC, power on and run).

② The biggest advantage of dual-channel CAN is the use of his relay function, if a dual-channel CAN without a repeater, then there is no difference with single-channel CAN.

③ CAN relay function: 1, to extend the receiving distance. 2. Signal isolation (to connect the two CAN bus that need to be isolated ). 3. The CAN network into the CAN repeater, without disconnecting the CAN network, to analyze the sending and receiving of the CAN network, as well as protocol analysis.

### 2.2 Transparent Transmission

The USB-CAN adapter and the **Supreme Edition of the CANalyst-II** analyzer provided by our company support the transparent transmission function. Transparent transmission: In the relay state, through the PC software, you can monitor the CAN1 and CAN2 data exchange in real time, and at the same time, you can also send data to the bus. The features are as follows:

① During relaying, you can also use the PC terminal software to monitor the device communication, realizing transparent transmission, and also send data to the bus. (At this time and USBCAN device operation and function is exactly the same, at the same time to realize the relay function, transparent transmission function.)

② Transparent transmission function enables detailed monitoring and analysis of data sending and receiving of individual nodes or buses without detaching from the bus. It provides convenience for protocol analysis and cracking, which cannot be

accomplished by other USBCAN devices.

More details of the functions need to be explored in the use of the device.

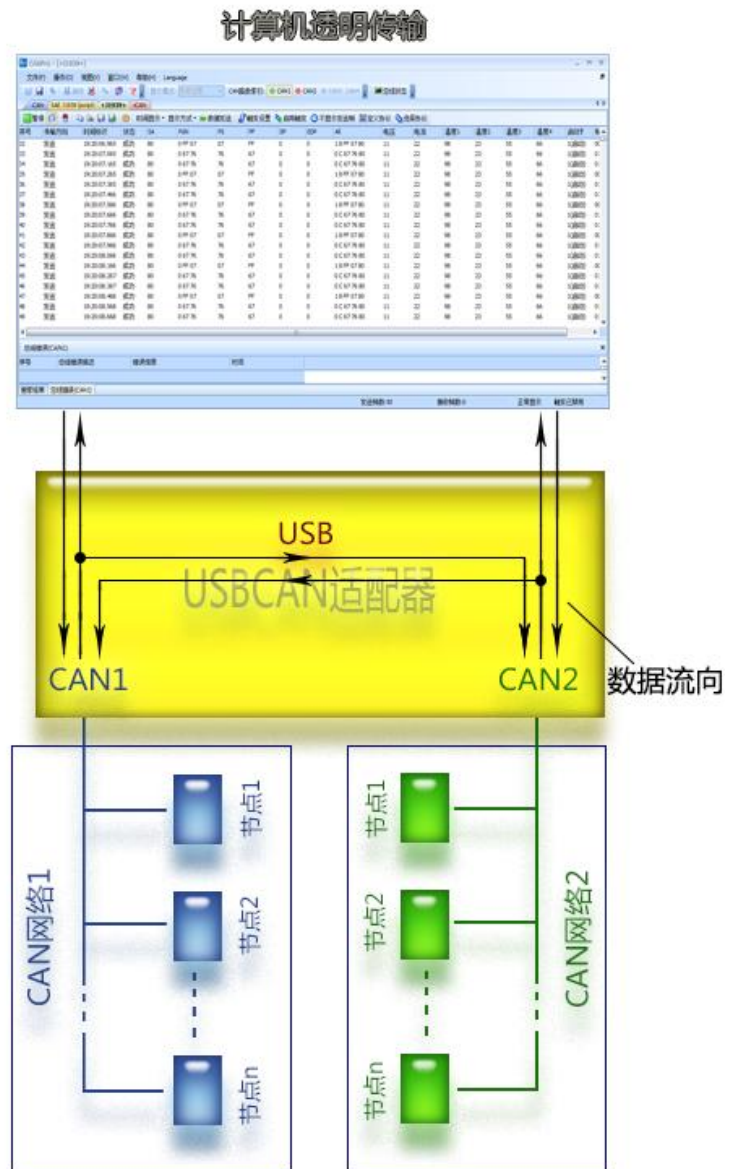
The following diagram shows the data flow of CAN relay and transparent transmission:

## 透明传输

中继状态下，通过PC机软件，可以实时监控CAN1、CAN2数据交换，同时还能向总线发送数据。

## 中继功能

无需PC机，CAN1，CAN2在硬件层实现数据交换。

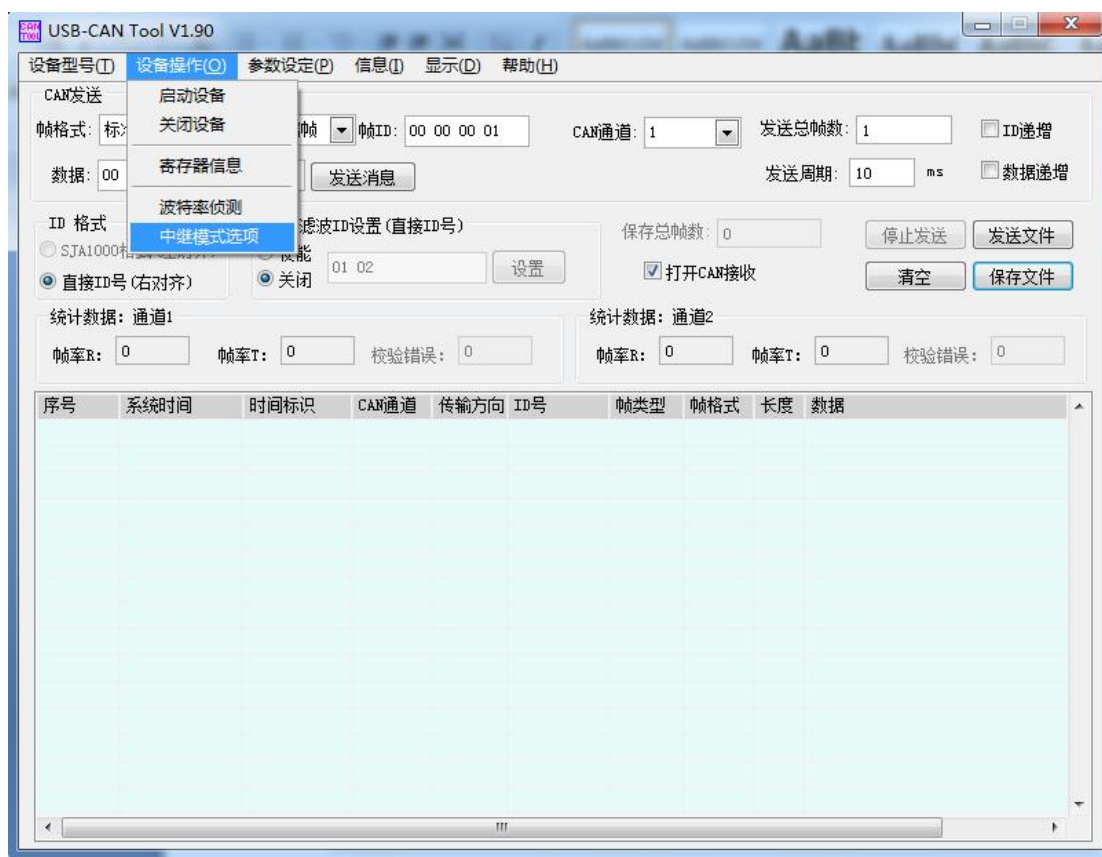


## Part III Relay Software User's Manual

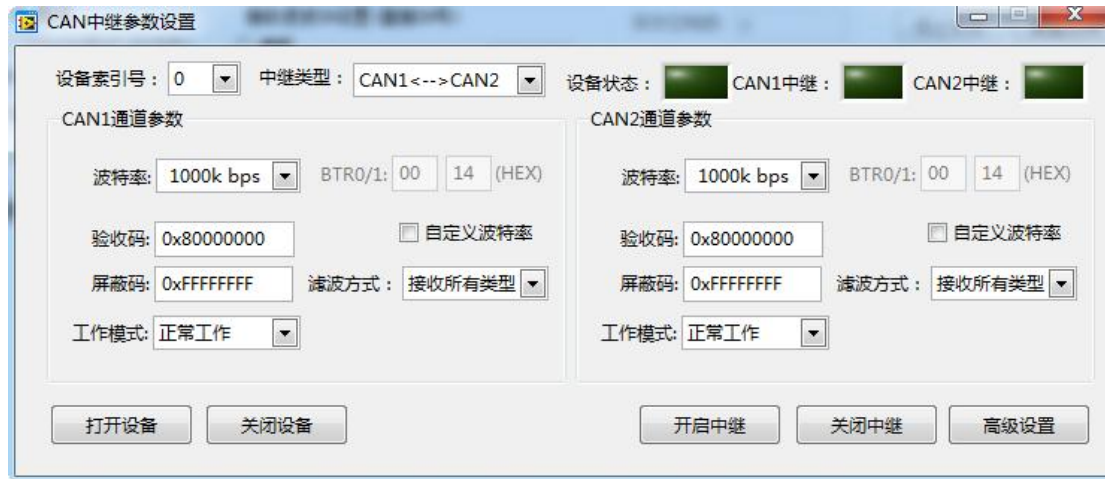
The relay software has been integrated into the USB CAN Tool software. For instructions on how to use the software, please refer to the relevant instruction manual of the USB CAN Tool.

### 3.1 How to open the relay software

Start USB-CAN Tool, click "Device Operation", "Relay Mode Option" menu item, you can open the relay plug-in.



The Relay Mode Configuration window is shown below:



## 3.2 CAN Relay Parameter Setting

Set the relay parameters in the CAN Relay Parameter Setting window opened via USB-CAN TOOL. These settings include: selecting the device index number, the relay type, the baud rate and working mode of CAN1 channel, and the baud rate and working mode of CAN2 channel, etc.

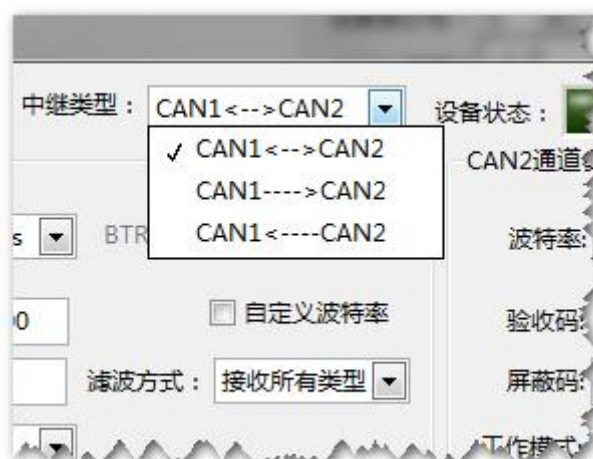
### 3.2.1 Device Index Number

Device index number, the sequence number of USB-CAN adapter in the computer. From 0, when the computer plugs in more than one USB CAN adapter, you need to select the device by the device index number. In most cases, there is only one USB CAN adapter in the computer, and it is sufficient to select 0 for the device index number.



### 3.2.2 Relay Types

The relay types include 3 types: CAN1 and CAN2 dual channels forwarding to each other, CAN1 forwarding to CAN2, and CAN2 forwarding to CAN1.



1、**(CAN1<-->CAN2)CAN1 and CAN2 dual-channel forwarding:** USB CAN adapter will receive data from CAN1 channel and send it through CAN2 channel at the same time; receive data from CAN2 channel and send it through CAN1 channel at the same time.

2、**(CAN1--->CAN2) CAN1 forwarding to CAN2:** USB CAN adapter will receive data from CAN1 channel and send it through CAN2 channel, but will not send data from CAN2 channel through CAN1 channel.

3、**(CAN1<---CAN2) CAN2 forwarding to CAN1:** USB CAN adapter will receive data from CAN2 channel and send it through CAN1 channel, but it will not



send data from CAN1 channel through CAN2 channel.

### 3.2.3 Baud Rate Setting

The baud rate setting includes two channels: CAN1 and CAN2. Each channel is set independently, and the available operations include regular baud rate setting, customized baud rate, filtering method, and working method.

These settings are basically the same as the CAN parameter settings in USB CAN TOOL, so we will not repeat here, please refer to the relevant manual of USB CAN Tool.

## 3.3 Operation Flow

The operation flow is shown in the following figure:

**Note:** Turn on the relay process: configure parameters - turn on the device - turn on the relay - turn off the device - Close the software.

Turn off the relay process: turn on the device - turn off the relay - turn off the device - turn off the software.



### 3.3.1 Selecting the Device Index Number

Select the serial number of the USB CAN adapter in the computer according to the serial number of the USB CAN adapter for which the relay function needs to be set. The serial number starts from 0. If only one adapter is connected in the computer,

the serial number is 0.

Serial number 1 is shown in the figure.

### 3.3.2 Selecting Relay Type

Select the relay type according to the site requirements.

Serial number 2 is shown in the figure.

### 3.3.3 Setting CAN1 Channel Parameters

Set the parameters of the bus connected to the CAN1 channel of the adapter, among which the baud rate parameter must match the existing baud rate of the bus, the filtering parameter can be set according to the actual needs, and the working mode can be selected as "Normal work".

Serial number 3 is shown in the figure.

### 3.3.4 Setting CAN2 Channel Parameters

Set the parameters of the bus connected to the CAN2 channel of the adapter, of which the baud rate parameter must match the existing baud rate of the bus, the filtering parameter can be set according to the actual needs, and the working mode can be selected as "normal operation".

Serial number 4 is shown in the figure.

**Note: The above 3.3.1~3.3.4 are 4 steps, the order can be interchanged, and it is not required to set up in the above order absolutely.**

### 3.3.5 Opening a Device

Click the Open Device button to open the selected device. After successful opening, the software will automatically read the relay on status of the device.

There are 3 "LED" indicators in the upper right corner of the interface:

**Device Status:** The light indicates that the device has been successfully opened;

**CAN1 relay:** The light indicates that the relay function of CAN1 channel has been turned on. (CAN1--->CAN2)

**CAN2 relay:** The light indicates that the relay function of CAN2 channel has been turned on. (CAN1<---CAN2)

**If both lights on at the same time, it means: (CAN1<-->CAN2).**

Serial number 5 is shown in the diagram.

### 3.3.6 Switching Relay On/Off

After the device has been successfully turned on, click on "Enable Relay" or "Close Relay" to turn on or off the relay function of the corresponding channel. If you click on the "Close relay" button, the relay function of both CAN1 and CAN2 channels will be closed.

Serial numbers 6 and 7 are shown in the figure.

### 3.3.7 Shutting Down the Device

After setting the relay function, click on the "Close Device" button to close the device.

Serial number 8 is shown in the figure.

### 3.3.8 Closing Software

To close the software, click the "X" button on the upper right corner of the window.

Serial number 9 is shown in the figure.

**Note:** After the relay function is configured, all the configuration parameters will be saved inside the adapter, and will be run directly next time when the power is turned on. At this time, all the configuration operations of the monitoring software are invalidated, only reading

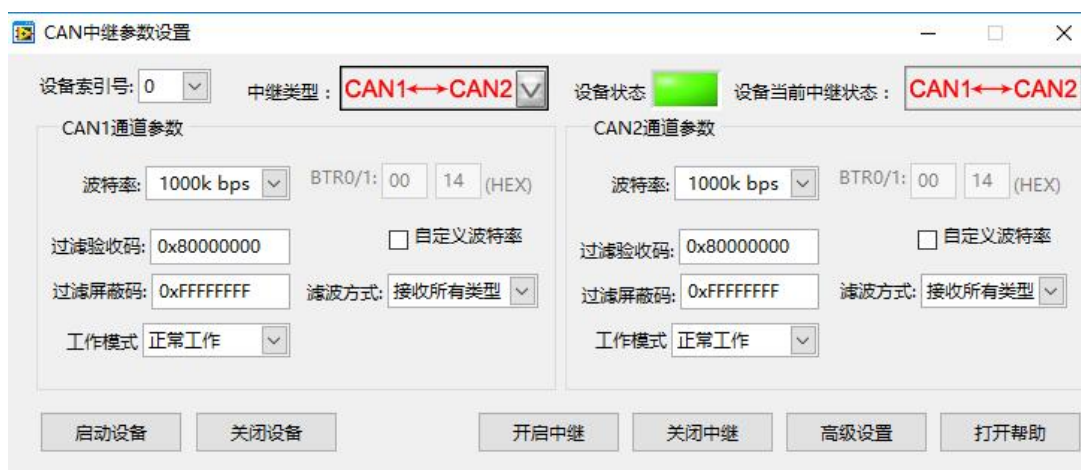
display or sending data. Therefore, when using as a debugging tool, you must turn off the CAN relay mode with the "CAN relay parameter setting software".

## Part IV Intelligent Filtering Function

The so-called intelligent filtering function means that the filter table can be flexibly configured to filter specific one or more IDs during relaying, and there is no limit to the number of IDs to be filtered. This function is performed with the aid of the CANtest software.

### 4.1 Enabling relaying

Enable the relay function according to the instructions in section 3. Note: The filter settings in the relay software are also valid.



Attention:

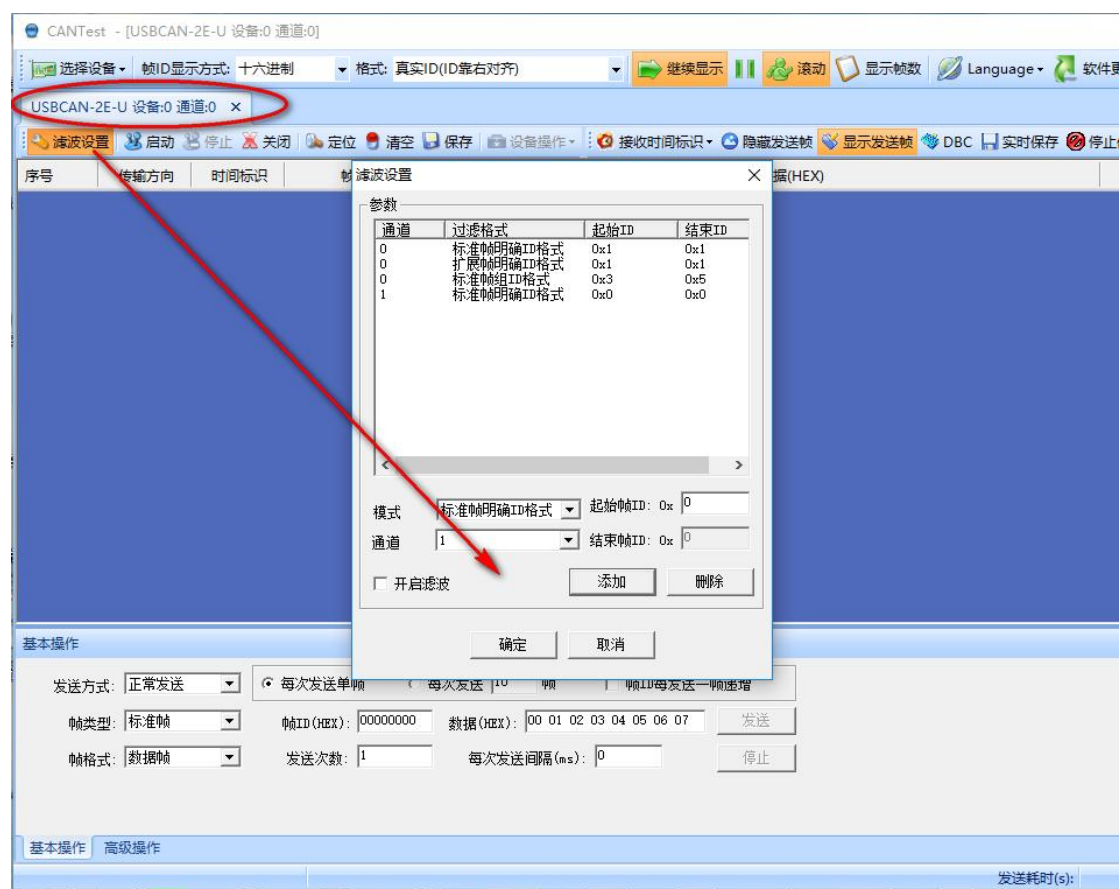
1. After turning on the relay, baud rate, filter acceptance code, filter blocking code and working mode, filtering mode parameters is locked. With other software to observe the data flow of the relay, the configuration parameters are invalid. When you need to modify, you can only use the relay software to turn off the relay function, and then re-open the relay.

2. The relay software filtering and intelligent filtering are two independent sets of filtering systems, only ID through two consecutive filtering will be forwarded. If intelligent filtering is used, it is recommended that in the relay software, the default configuration is used, that is, all data passes.

## 4.2 Configuring Smart Filtering Parameters

Reference: CD-ROM Data \ Debugging Tools \ Zhou Ligong ZLG Debugging Tools \ How to Compatible Use Zhou Ligong Related Software \ 1. How to Compatible Use Zhou Ligong CANTest Software.pdf, install CANTest and replace the corresponding DLL file.

Start CANtest, select the device "USBCAN-2E-U", other parameters can be defaulted. Click "Filter Settings".



Filtering instructions:

- 1、ID or ID group in the filter table, represents IDs that need to be relayed and forwarded. IDs not in this range are not relayed and forwarded.
- 2、Channel number 0 indicates the ID configuration received by CAN1 and relayed to CAN2 (provided that the relay function in the direction of CAN1->CAN2 is enabled)

3、 Channel No. 1 indicates the ID configuration of receiving from CAN2 and relaying to CAN1 (provided that the relay function in the direction of CAN2->CAN1 is enabled).

4、 Enabling filtering needs to be checked, and then tap OK. The whole filter table is only enabled.